

Reinforcement Learning

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Agenda

pure nature

- Markov Processes (state, no reward, no action)

human can set score or reward

- Markov Reward Processes (state, reward, no action)
- Markov Decision Processes (state, reward, action)

how healthy, score, action you take

- Extensions beyond MDPs

Example: plant watermelons



Healthy



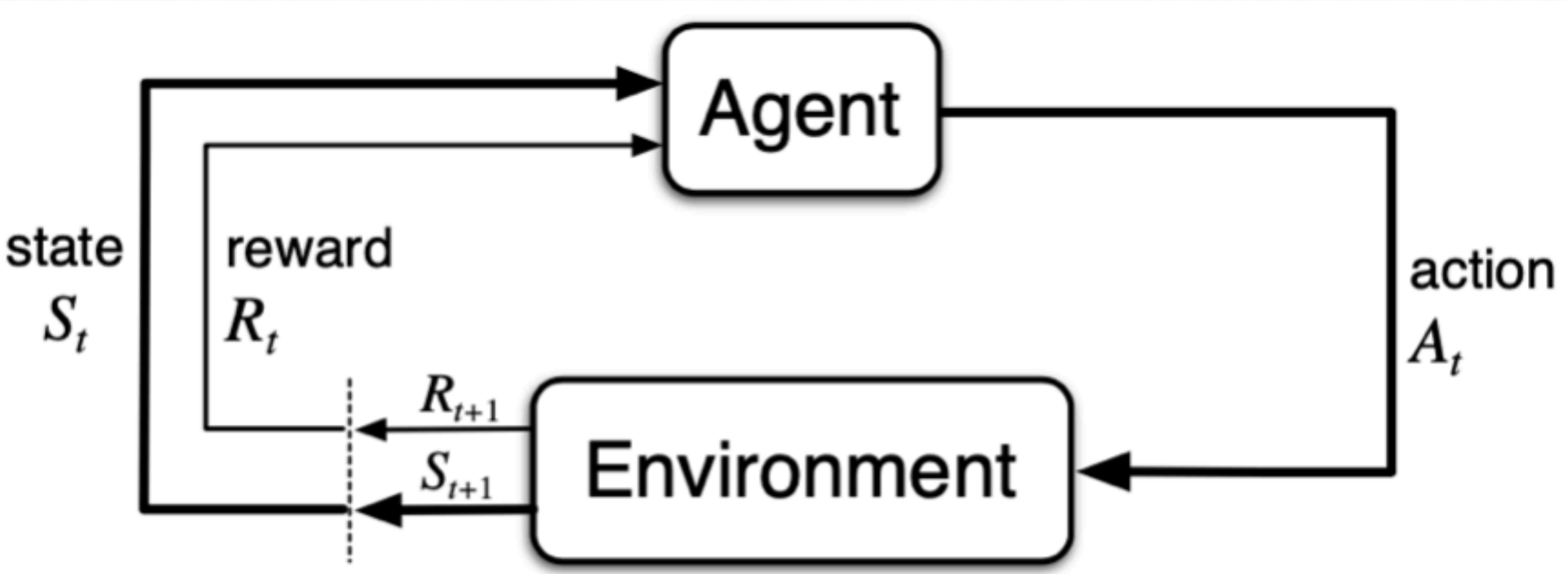
Needing help

- MP: let nature take its own course, no interference (e.g., states: green/yellow leaf)
- MRP: states, and rewards [r(green)=1, r(yellow)=-1]
- MDP: states, reward, actions (yellow: water; green: nothing)

d secrete离散，finite有限 number of state

Review on MDP

- Formally describes an environment for RL
- Environment: fully observable
- Current state completely characterises the process



The agent-environment interaction in a MDP

Almost all RL problems can be formalised as MDP's

- Optimal control primarily deals with continuous MDPs
- Partially observable problems can be converted into MDPs (=MDP where state is history)
- Bandits are MDPs with one state

Markov Process

(Markov Chain)

Markov Property (Memoryless)

- Definition: A state S_t is Markov iff
- $$\mathbf{P}(S_{t+1} \mid S_t) = \mathbf{P}(S_{t+1} \mid S_1, \ldots, S_t)$$
- i.e., Future is independent of past given present

State Transition Matrix

- For a finite number of states (n),
-

$$P = \begin{pmatrix} P_{11} & P_{12} & \cdots & P_{1n} \\ P_{21} & P_{22} & \cdots & P_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ P_{n1} & P_{n2} & \cdots & P_{nn} \end{pmatrix} = \begin{pmatrix} p(1|1) & p(2|1) & \cdots & p(n|1) \\ p(1|2) & p(2|2) & \cdots & p(n|2) \\ \vdots & \vdots & \ddots & \vdots \\ p(1|n) & p(2|n) & \cdots & p(n|n) \end{pmatrix}$$

- Note: no rewards, no actions

Markov process

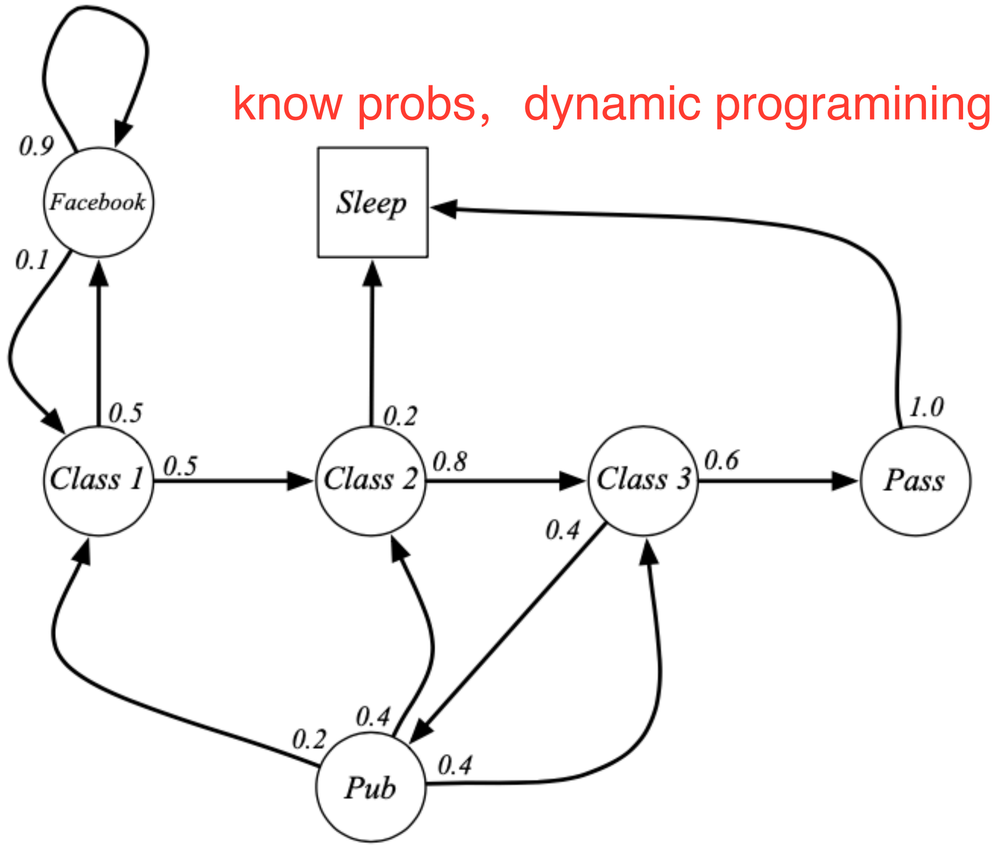
- A Markov Process (Chain) is a tuple $\langle \mathcal{S}, \mathcal{P} \rangle$
  - $\mathcal{S}$ is a (finite) set of states
  - $\mathcal{P}$ is a state transition/dynamics probability matrix,

$$p(s'|s) := P_{ss'} := \mathbf{P}(S_{t+1} = s'|S_t = s)$$

Example: Student attending class

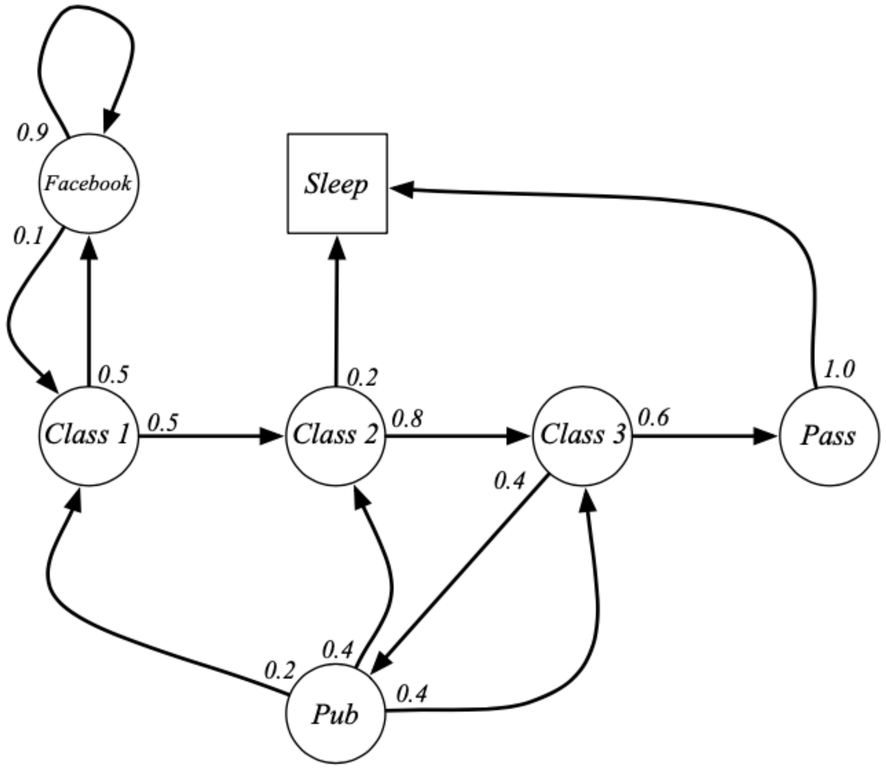
learning from begining，may not know probs but can estimate from history

know probs，dynamic programining，not know，RL



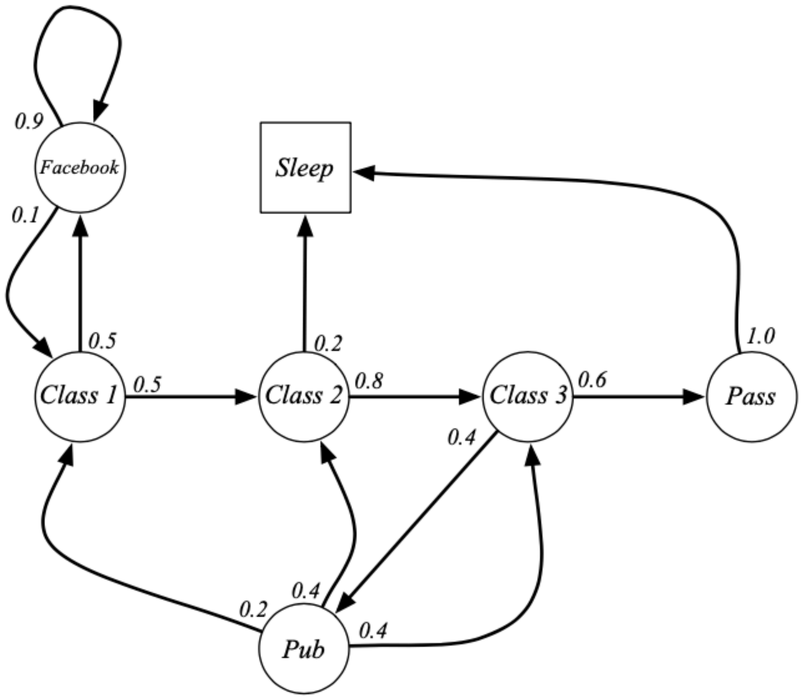
- 7 states:
- C1, C2, C3, Pass, Sleep, FB, Pub

Transition matrix



$$\mathcal{P} = \begin{array}{c} \\ C1 \\ C2 \\ C3 \\ Pass \\ Pub \\ FB \\ Sleep \end{array} \begin{array}{c} \begin{array}{ccccccc} C1 & C2 & C3 & Pass & Pub & FB & Sleep \end{array} \\ \begin{bmatrix} & 0.5 & & & & 0.5 & \\ & & 0.8 & & & & 0.2 \\ & & & 0.6 & 0.4 & & \\ & & & & & & 1.0 \\ 0.2 & 0.4 & 0.4 & & & & \\ 0.1 & & & & & 0.9 & \\ & & & & & & 1 \end{bmatrix} \end{array}$$

Sample episodes: initial value $S_1 = C1$



- C1 C2 C3 Pass Sleep
- C1 FB FB C1 C2 Sleep
- C1 C2 C3 Pub C2 C3 Pass Sleep
- C1 FB FB C1 C2 C3 Pub C1 FB FB FB C1 C2 C3 Pub C2 Sleep

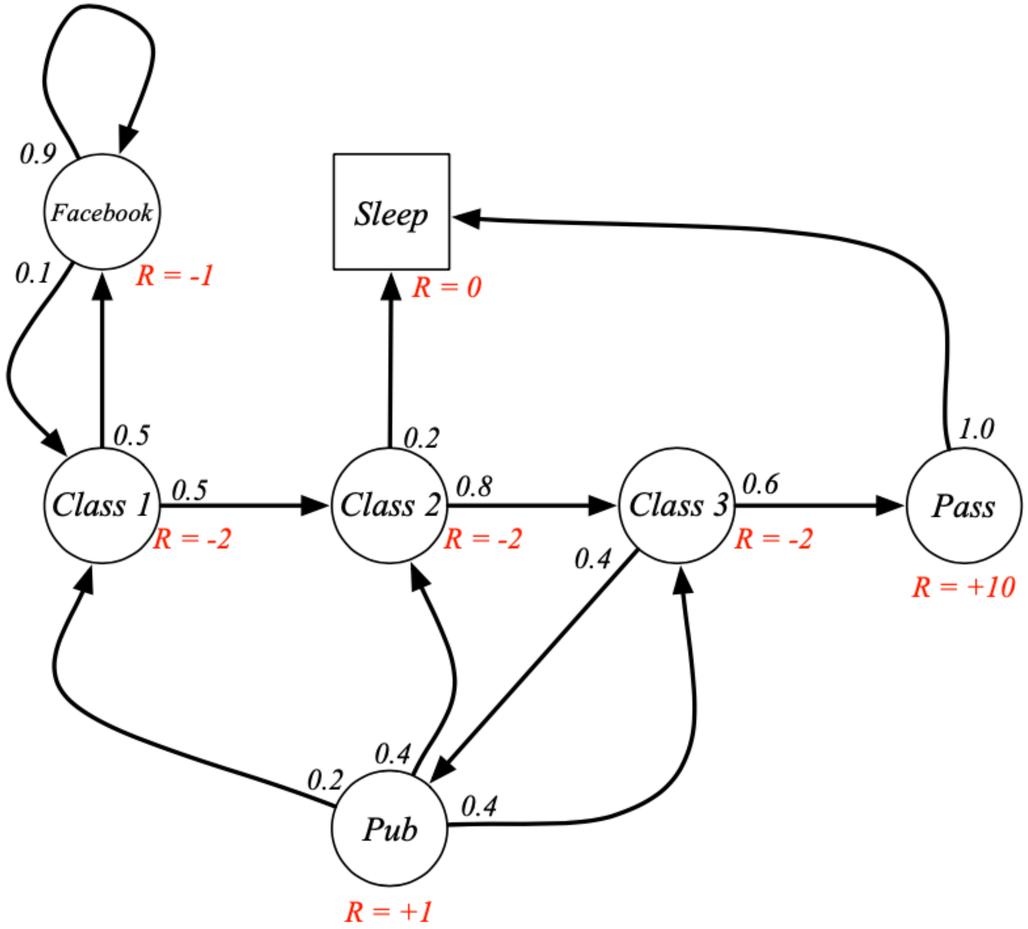
Markov Reward Process

(Markov Process + Reward)

Markov Reward Process

- A Markov Reward Process (MRP) is a tuple $\langle \mathcal{S}, \mathcal{P}, \mathcal{R}, \gamma \rangle$
  - $\mathcal{S}$ is a (finite) set of states
  - $\mathcal{P}$ is a transition probability matrix
  - $$p(s'|s) := P_{ss'} := \mathbf{P}(S_{t+1} = s' | S_t = s)$$
  - $\mathcal{R}$ is a reward function (or vector)
  - $$r_s := \mathbf{E}(R_t | S_t = s)$$ different time may have different reward
  - $\gamma \in [0,1]$ is a discount factor

Example: Student MRP



Reward (immediate)

- R(C1)=R(C2)=R(C3)=-2
- R(Pass)=10, R(Sleep)=0
- R(FB)=-1, R(Pub)=1

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Q: how to assign reward ?

Return = total discounted reward

- Return G_t is the total discounted reward from time-step t:

$$G_t := R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+2} + \ldots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

- The discount $\gamma \in [0, 1]$ is the present value of future rewards
- The value of receiving reward R after $k + 1$ time-steps is $\gamma^k R$
- γ close to 0: “myopic” evaluation

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- γ close to 1: “far-sighted” evaluation

how fulture reward looks like (0.9999 cloose to 1 not equal to 1)

Q: how to choose γ?

Why discount?

- Mathematically convenient to discount rewards
- Avoids infinite returns in cyclic Markov processes
- Uncertainty about the future may not be fully represented
- If the reward is financial, future earnings should be discounted
- Animal/human behaviour shows preference for immediate reward
- Sometimes possible to choose $\gamma = 1$, e.g. if all sequences terminate.

如果G能被有限t确定就有upper bound

r discount avoid doing same thing to get reward，no exploration

discount/ uncertaining

for finite state，if no converge problem

Value Function

- The state value function $v(s)$ is the expected return starting from state s

try to find action from bad state to good state

$$v(s) := \mathbf{E}[G_t \mid S_t = s]$$

  - i.e. $v(s)$ = long-term value of state s

Example: Student MRP Returns

Starting from $S_1 = C1$ with $\gamma = 1/2$

$$G_t := R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+2} + \dots$$

Sample	Sample returns
C1 C2 C3 Pass Sleep	$v_1 = -2 - 2 * \frac{1}{2} - 2 * \frac{1}{4} + 10 * \frac{1}{8} = -2.25$
C1 FB FB C1 C2 Sleep	$v_1 = -2 - 1 * \frac{1}{2} - 1 * \frac{1}{4} - 2 * \frac{1}{8} - 2 * \frac{1}{16} = -3.125$
C1 C2 C3 Pub C2 C3 Pass Sleep	$v_1 = -2 - 2 * \frac{1}{2} - 2 * \frac{1}{4} + 1 * \frac{1}{8} - 2 * \frac{1}{16} \dots = -3.41$
C1 FB FB C1 C2 C3 Pub C1 ...	$v_1 = -2 - 1 * \frac{1}{2} - 1 * \frac{1}{4} - 2 * \frac{1}{8} - 2 * \frac{1}{16} \dots = -3.20$
FB FB FB C1 C2 C3 Pub C2 Sleep	

Finding $v(C1)$ by Monte Carlo

$v(C1) \approx$ Average of sample returns

Sample

C1 C2 C3 Pass Sleep

C1 FB FB C1 C2 Sleep

C1 C2 C3 Pub C2 C3 Pass Sleep

C1 FB FB C1 C2 C3 Pub C1 ...

FB FB FB C1 C2 C3 Pub C2 Sleep

Sample returns

$$v_1 = -2 - 2 * \frac{1}{2} - 2 * \frac{1}{4} + 10 * \frac{1}{8} = -2.25$$

$$v_1 = -2 - 1 * \frac{1}{2} - 1 * \frac{1}{4} - 2 * \frac{1}{8} - 2 * \frac{1}{16} = -3.125$$

$$v_1 = -2 - 2 * \frac{1}{2} - 2 * \frac{1}{4} + 1 * \frac{1}{8} - 2 * \frac{1}{16} \ldots = -3.41$$

$$v_1 = -2 - 1 * \frac{1}{2} - 1 * \frac{1}{4} - 2 * \frac{1}{8} - 2 * \frac{1}{16} \ldots = -3.20$$

Example: Student MRP Returns

Starting from $S_1 = C1$ with $\gamma = 1/2$

$$G_t := R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+2} + \dots$$

Q: all negative returns?



Sample	Sample returns		
C1 C2 C3 Pass Sleep	$v_1 = -2 - 2 * \frac{1}{2} - 2 * \frac{1}{4} + 10 * \frac{1}{8}$	=	-2.25
C1 FB FB C1 C2 Sleep	$v_1 = -2 - 1 * \frac{1}{2} - 1 * \frac{1}{4} - 2 * \frac{1}{8} - 2 * \frac{1}{16}$	=	-3.125
C1 C2 C3 Pub C2 C3 Pass Sleep	$v_1 = -2 - 2 * \frac{1}{2} - 2 * \frac{1}{4} + 1 * \frac{1}{8} - 2 * \frac{1}{16} \dots$	=	-3.41
C1 FB FB C1 C2 C3 Pub C1 ...	$v_1 = -2 - 1 * \frac{1}{2} - 1 * \frac{1}{4} - 2 * \frac{1}{8} - 2 * \frac{1}{16} \dots$	=	-3.20
FB FB FB C1 C2 C3 Pub C2 Sleep			

Example: Student MRP Returns

Starting from $S_1 = C1$ with $\gamma = 1/2$

What if $\gamma = 1$?

$$G_t := R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+2} + \dots$$

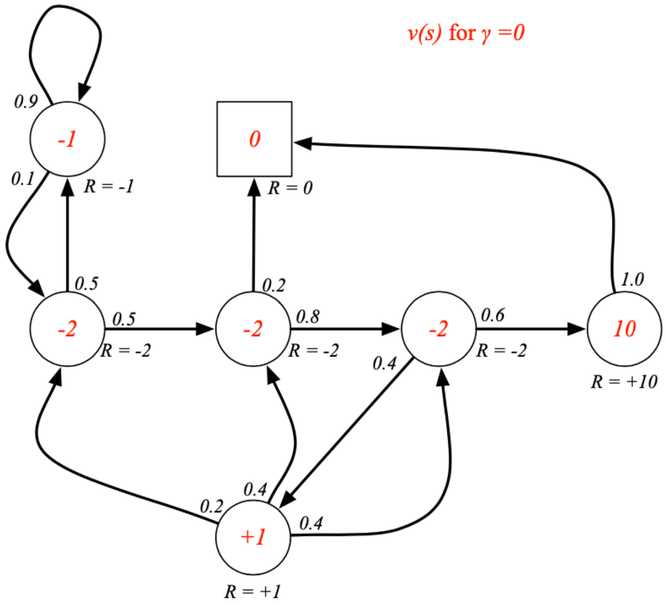
Q: all negative returns?

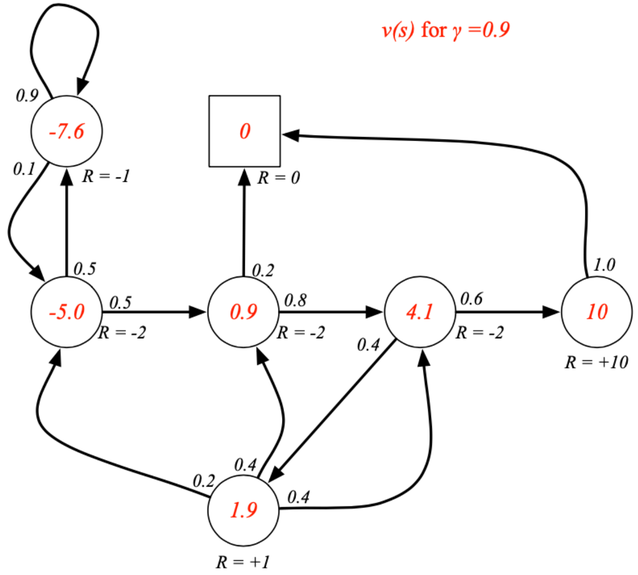
Sample	Sample returns		
C1 C2 C3 Pass Sleep	$v_1 = -2 - 2 * \frac{1}{2} - 2 * \frac{1}{4} + 10 * \frac{1}{8}$	=	-2.25
C1 FB FB C1 C2 Sleep	$v_1 = -2 - 1 * \frac{1}{2} - 1 * \frac{1}{4} - 2 * \frac{1}{8} - 2 * \frac{1}{16}$	=	-3.125
C1 C2 C3 Pub C2 C3 Pass Sleep	$v_1 = -2 - 2 * \frac{1}{2} - 2 * \frac{1}{4} + 1 * \frac{1}{8} - 2 * \frac{1}{16} \dots$	=	-3.41
C1 FB FB C1 C2 C3 Pub C1 ...	$v_1 = -2 - 1 * \frac{1}{2} - 1 * \frac{1}{4} - 2 * \frac{1}{8} - 2 * \frac{1}{16} \dots$	=	-3.20
FB FB FB C1 C2 C3 Pub C2 Sleep			

Example: State value function for Student MRP

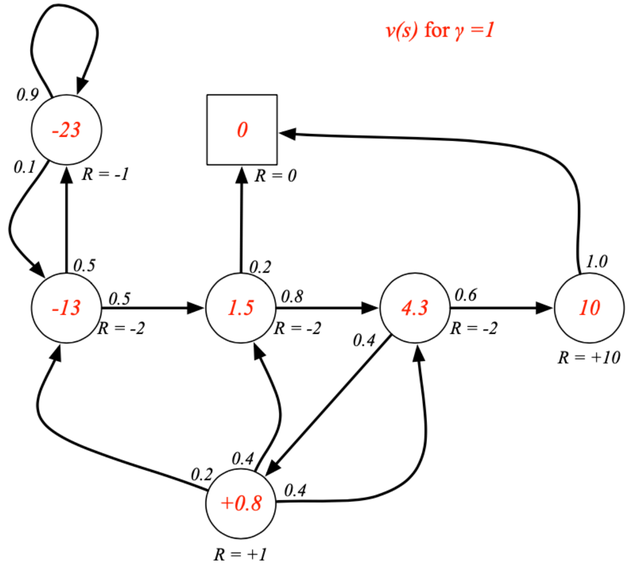
Starting from $S_1 = C1$

do not know fulture



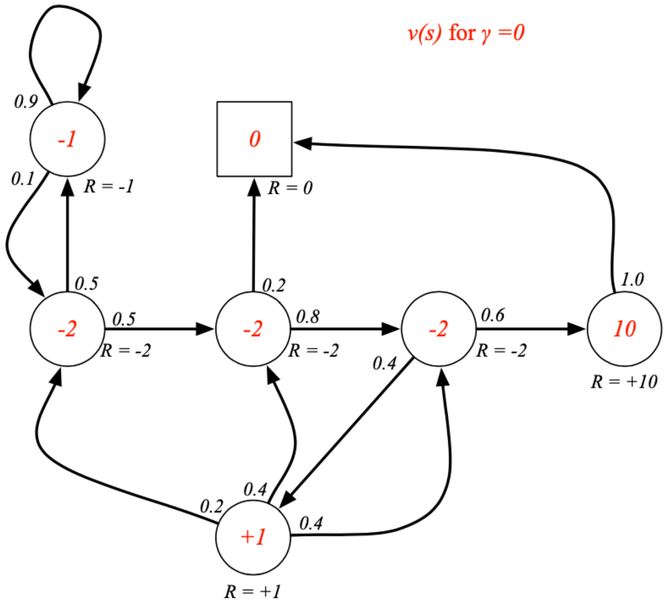


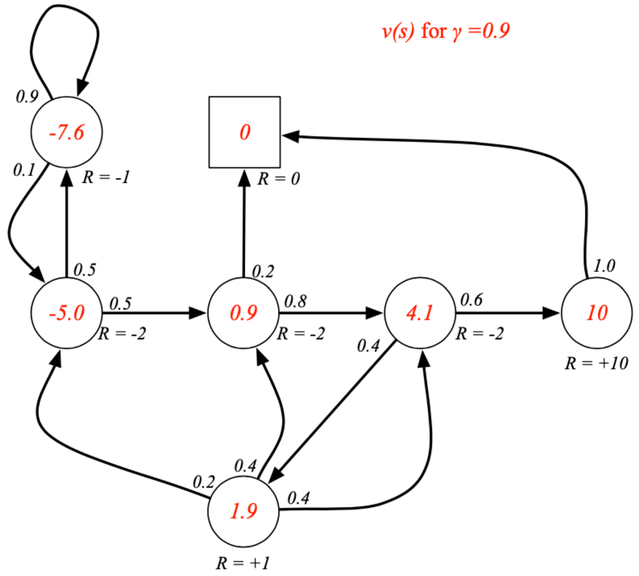
如果所有的都可以observe?

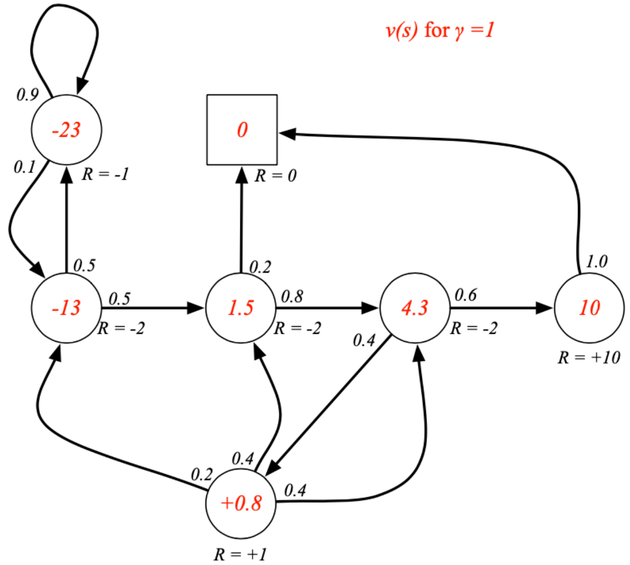


Example: State value function for Student MRP

Starting from $S_1 = C1$







Far-sighted: More effort in the beginning, soon getting better

Finding $v(s)$ by Bellman equation

Bellman equation for MRPs

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+2} + \dots$$

$$= R_{t+1} + \gamma(R_{t+2} + \gamma R_{t+2} + \dots)$$

$$= R_{t+1} + \gamma G_{t+1}$$

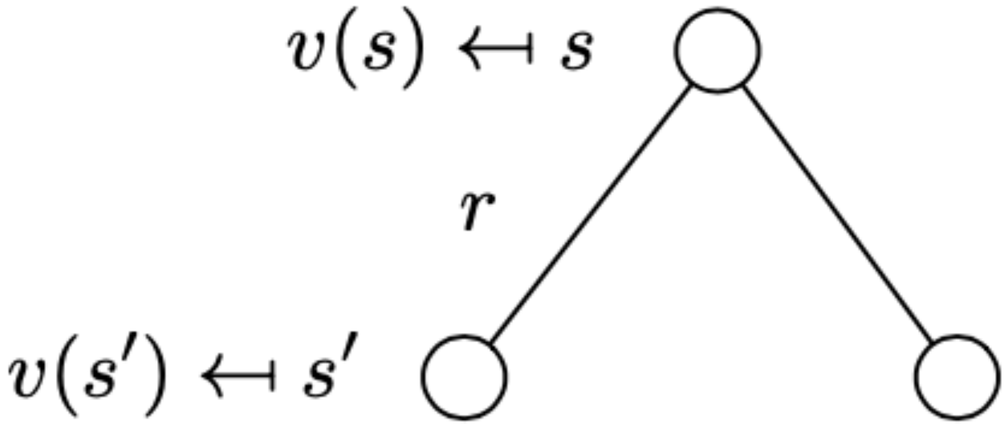
today + fulture

Taking expectation

$$
\begin{aligned}
v(s) &:= \mathbf{E}[G_t \mid S_t = s] \\
&= \mathbf{E}[R_{t+1} + \gamma G_{t+1} \mid S_t = s] \\
&= \mathbf{E}[R_{t+1} \mid S_t = s] + \gamma \mathbf{E}[G_{t+1} \mid S_t = s] \\
&= \mathbf{E}[R_{t+1} \mid S_t = s] + \gamma \mathbf{E}[v(S_{t+1}) \mid S_t = s] \\
&= \mathscr{R}_s + \gamma \sum_{s' \in \mathcal{S}} \mathscr{P}_{ss'} v(s') \\
&= \mathbf{E}[R_{t+1} + \gamma v(S_{t+1}) \mid S_t = s]
\end{aligned}
$$

$$v(s) = \mathbb{E}\left[R_{t+1} + \gamma v(S_{t+1}) \mid S_t = s\right]$$

one step sampling link v

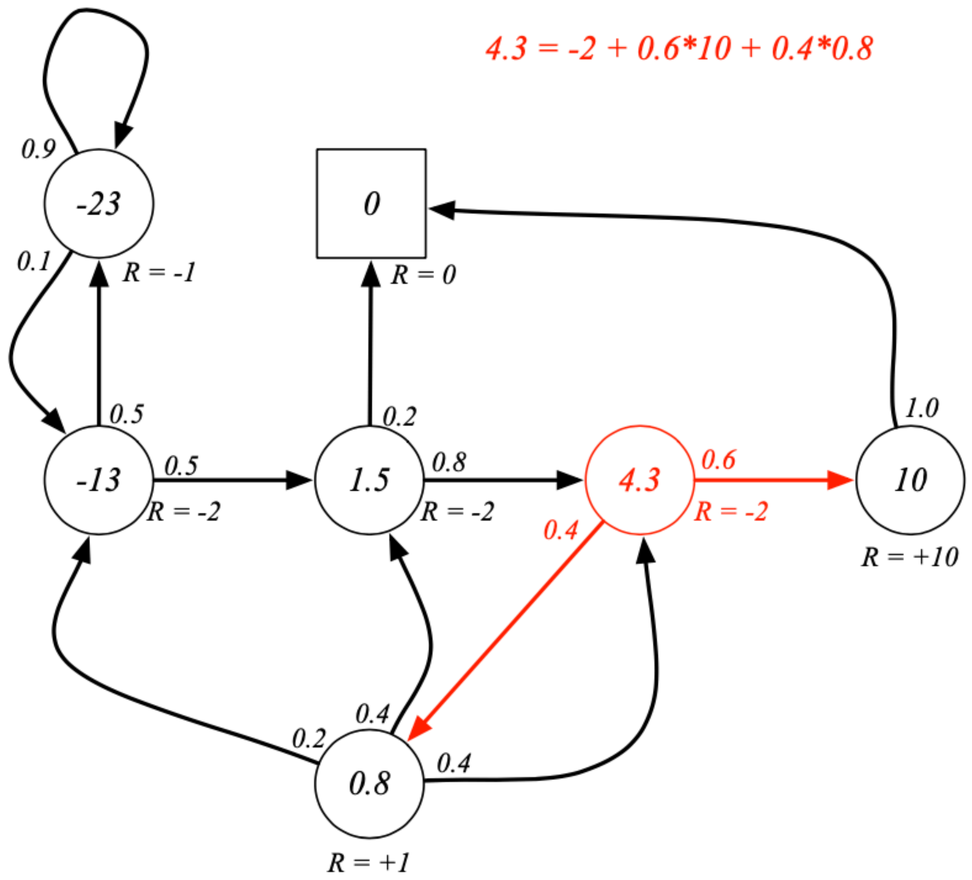


R已知，v需要计算

$$v(s) = \mathcal{R}_s + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'} v(s')$$

- The value function =
- immediate reward + discounted value of successor state

Example: Bellman equation for Student MRP



$$v(s) = \mathcal{R}_s + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'} v(s') \qquad \gamma = 1$$

e.g.

$$v(C3) = R(C3) + P_{C3,Pass} v(Pass) + P_{C3,Pub} v(Pub)$$

$$4.3 = -2 + 0.6 * 10 + 0.4 * 0.8$$

Solving Bellman equation

The Bellman equation can be expressed concisely using matrices,

$$v = \mathcal{R} + \gamma \mathcal{P} v$$

where v is a column vector with one entry per state

$$\begin{bmatrix} v(1) \\ \vdots \\ v(n) \end{bmatrix} = \begin{bmatrix} \mathcal{R}_1 \\ \vdots \\ \mathcal{R}_n \end{bmatrix} + \gamma \begin{bmatrix} \mathcal{P}_{11} & \dots & \mathcal{P}_{1n} \\ \vdots & & \\ \mathcal{P}_{11} & \dots & \mathcal{P}_{nn} \end{bmatrix} \begin{bmatrix} v(1) \\ \vdots \\ v(n) \end{bmatrix}$$

Directly solving Bellman equation

- The Bellman equation is a linear equation
- It can be solved directly:

r 小于1的时候可逆，等于1不太确定

$$v = \mathcal{R} + \gamma \mathcal{P} v$$

$$(I - \gamma \mathcal{P})\, v = \mathcal{R}$$

$$v = (I - \gamma \mathcal{P})^{-1} \mathcal{R}$$

Computational complexity

- Direct solution: $O(N^3)$ for N states
- e.g., if N=1000, $N^3 = 1{,}000{,}000{,}000$
- Only possible for small MRPs

Alternatives: Iterative methods

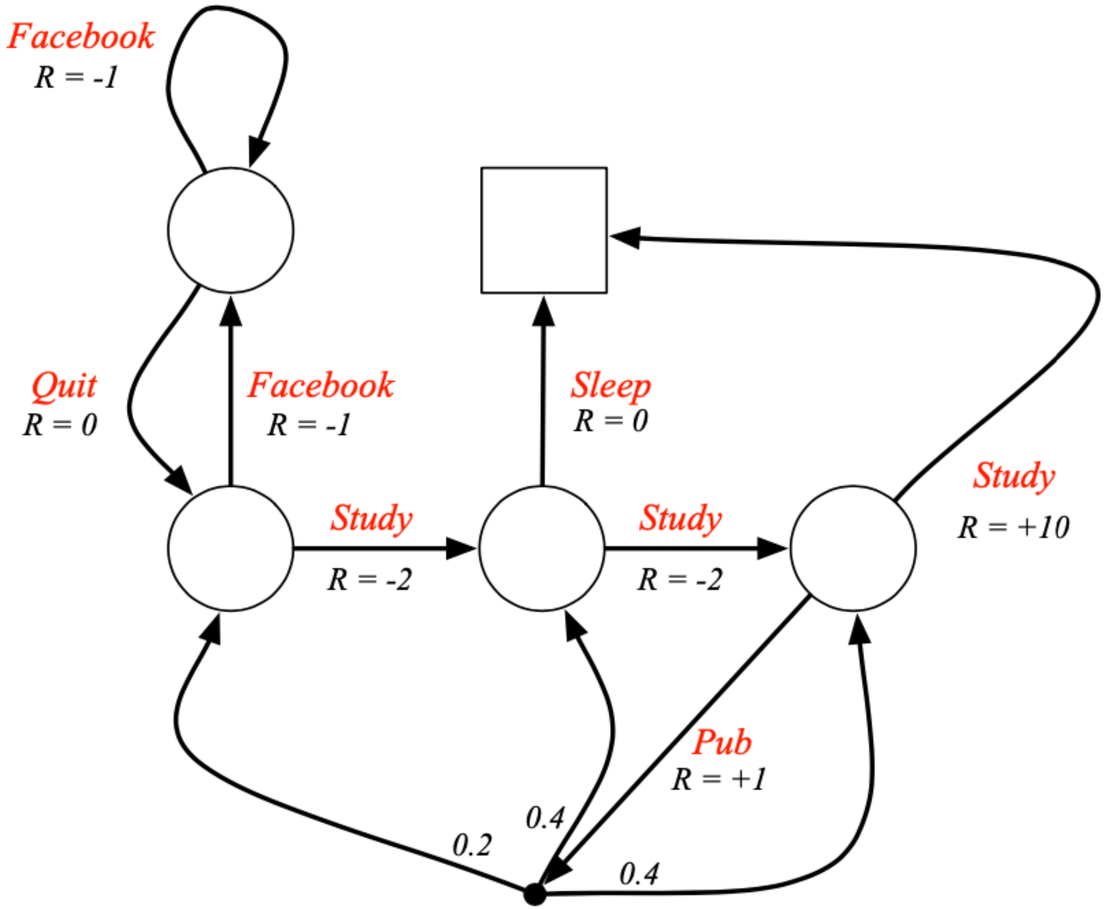
- Dynamic programming
- Monte-Carlo evaluation
- Temporal-Difference learning

Markov Decision Process

Definition

- A Markov Decision Process (MDP) is a tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$
  - $\mathcal{S}$ is a (finite) set of states
  - $\mathcal{A}$ is a (finite) set of actions
  - $\mathcal{P}$ is a transition probability matrix
  - $$P^a_{ss'} := \mathbf{P}(S_{t+1} = s' | S_t = s, A_t = a)$$
  - $\mathcal{R}$ is a reward function (or vector)
  - $$R^a_s := \mathbf{E}(R_t | S_t = s, A_t = a)$$
  - $\gamma \in [0,1]$ is a discount factor

Example: Student MDP



- States: C1, C2, C3, FB, End
- Actions: Study, FB, Pub, Quit, Sleep

Policy

- A policy is a distribution over actions given states

$$\pi(a \mid s) = \mathbf{P}[A_t = a \mid S_t = s]$$

  - π fully defines the behaviour of an agent
  - MDP policies depend on the current state (not history)
  - Assumption: Policy is stationary: $A_t \sim \pi(\cdot \mid S_t)$

pai 是跟t无关的，下标没有t

Given an MDP $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ and a policy π

- $S_1, S_2, \ldots$ is a Markov process $\langle \mathcal{S}, \mathcal{P}^\pi \rangle$

a larger matrix sa times s

- $$p_\pi(s' \mid s) := P^\pi_{s,s'} = \sum_{a \in A} \pi(a \mid s) P^a_{ss'}$$

action 和t无关，不管state1是一开始就在s1还是绕了一圈回到s1

- $S_1, R_2, S_2, R_3, \ldots$ is an MRP $\langle S, P^\pi, R^\pi, \gamma \rangle$

- $$r_\pi(s) := R^\pi_s = \sum_{a \in A} \pi(a \mid s) R^a_s$$

Value function

- The state-value function $v_\pi(s)$ of an MDP is the expected return starting from state s, and then following policy π

$$v_\pi(s) = \mathbf{E}_\pi[G_t \mid S_t = s]$$

if a is chonen based on policy v q is equal

- The action-value function $q_\pi(s,a)$ is the expected return starting from state s, taking action a, and then following policy π

Q leaening, find action

$$q_\pi(s,a) = \mathbf{E}_\pi[G_t \mid S_t = s, A_t = a]$$

Bellman equation for MDPs

The state-value function can again be decomposed into immediate reward plus discounted value of successor state,

$$v_\pi(s) = \mathbb{E}_\pi \left[R_{t+1} + \gamma v_\pi(S_{t+1}) \mid S_t = s \right]$$

The action-value function can similarly be decomposed,

$$q_\pi(s, a) = \mathbb{E}_\pi \left[R_{t+1} + \gamma q_\pi(S_{t+1}, A_{t+1}) \mid S_t = s, A_t = a \right]$$

Derivation of $v_\pi(s)$

$$
\begin{aligned}
v_\pi(s) &:= \mathbf{E}_\pi[G_t \mid S_t = s] \\
&= \mathbf{E}_\pi[R_{t+1} + \gamma G_{t+1} \mid S_t = s] \\
&= \mathbf{E}_\pi[R_{t+1} \mid S_t = s] + \gamma \mathbf{E}_\pi[G_{t+1} \mid S_t = s] \\
&= \mathbf{E}_\pi[R_{t+1} \mid S_t = s] + \gamma \mathbf{E}_\pi[v(S_{t+1}) \mid S_t = s] \qquad (*) \\
&= \mathscr{R}_s^\pi + \gamma \sum_{s' \in \mathcal{S}} \mathscr{P}_{ss'}^\pi v_\pi(s') \\
&= \mathbf{E}_\pi[R_{t+1} + \gamma v_\pi(S_{t+1}) \mid S_t = s]
\end{aligned}
$$

Derivation of (*)

$$\mathbf{E}_\pi[G_{t+1} \mid S_t = s] = \sum_{s'} \mathbf{E}_\pi[G_{t+1} \mid S_t = s, S_{t+1} = s'] \; P_{ss'}$$

$$= \sum_{s'} \mathbf{E}_\pi[G_{t+1} \mid S_{t+1} = s'] \; P_{ss'}$$

$$= \sum_{s'} \mathbf{E}_\pi[G_{t+1} \mid S_{t+1} = s'] \; P_{ss'}$$

？ 积分积掉了

$$= \mathbf{E}_\pi[v(S_{t+1}) \mid S_t = s]$$

Derivation of $q_\pi(s, a)$

$$
\begin{aligned}
q_\pi(s, a) &:= \mathbf{E}_\pi[G_t \,|\, S_t = s, A_t = a] \\
&= \mathbf{E}_\pi[R_{t+1} + \gamma G_{t+1} \,|\, S_t = s, A_t = a] \\
&= \mathbf{E}_\pi[R_{t+1} \,|\, S_t = s, A_t = a] + \gamma \mathbf{E}_\pi[G_{t+1} \,|\, S_t = s, A_t = a] \\
&= \mathbf{E}_\pi[R_{t+1} \,|\, S_t = s, A_t = a] + \gamma \mathbf{E}_\pi[q(S_{t+1}, A_{t+1}) \,|\, S_t = s, A_t = a] \\
&= \mathbf{E}_\pi[R_{t+1} + \gamma q_\pi(S_{t+1}, A_{t+1}) \,|\, S_t = s, A_t = a]
\end{aligned}
$$

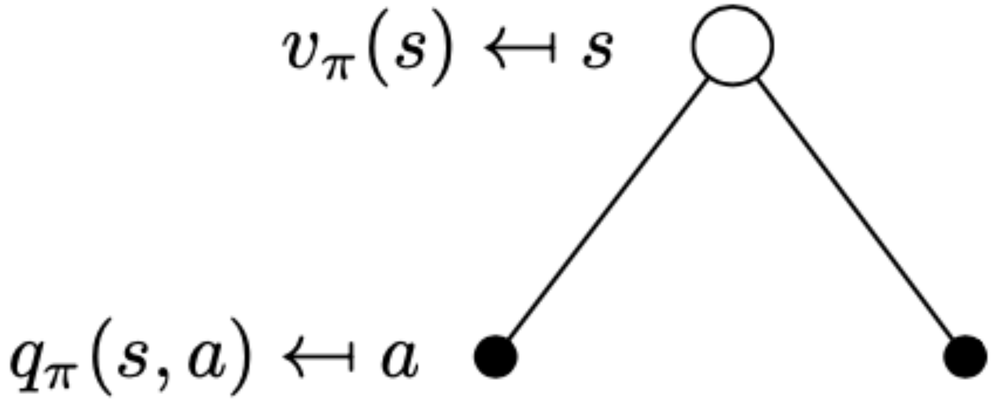
Again

- The value function =
- immediate reward + discounted value of successor state

$$v_\pi(s) = \mathbf{E}_\pi[R_{t+1} + \gamma v_\pi(S_{t+1}) \,|\, S_t = s]$$

$$q_\pi(s,a) = \mathbf{E}_\pi[R_{t+1} + \gamma q_\pi(S_{t+1}, A_{t+1}) \,|\, S_t = s, A_t = a]$$

Bellman Expectation Equation for v_π



$$v_\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) q_\pi(s,a)$$

(v_π depends on q_π)

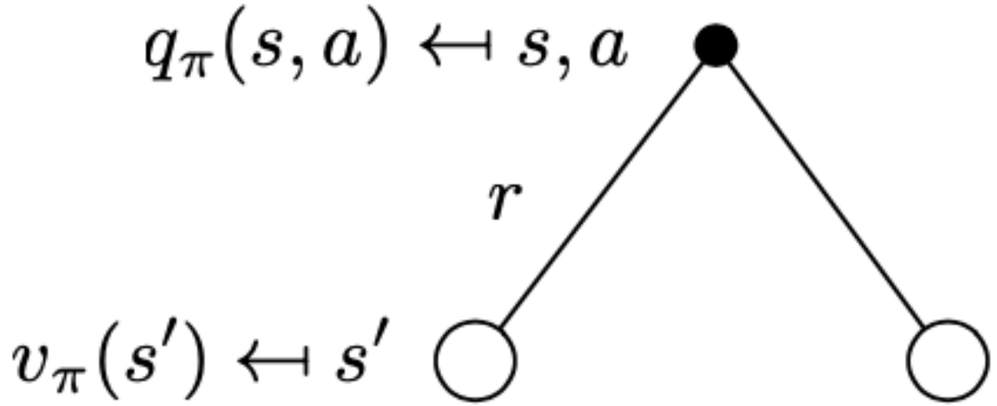
Derivation

$$v_\pi(s) := \mathbf{E}_\pi[G_t \,|\, S_t = s]$$

$$= \sum_{a \in \mathcal{A}} \pi(a \,|\, s)\ \mathbf{E}_\pi[G_t \,|\, S_t = s, A_t = a]$$

$$= \sum_{a \in \mathcal{A}} \pi(a \,|\, s)\ q_\pi(s, a)$$

Bellman Expectation Equation for q_π



$$q_\pi(s,a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a v_\pi(s')$$

(q_π depends on v_π)

Derivation

$$q_\pi(s,a) := \mathbf{E}_\pi[G_t \,|\, S_t = s, A_t = a]$$

$$= \mathbf{E}_\pi[R_{t+1} \,|\, S_t = s, A_t = a] + \gamma \mathbf{E}_\pi[G_{t+1} \,|\, S_t = s, A_t = a]$$

改写成s'的条件概率

$$= r_\pi(s,a) + \gamma \sum_{s' \in \mathcal{S}} p(s' \,|\, s,a)\ \mathbf{E}_\pi[G_{t+1} \,|\, S_t = s, A_t = a, S_{t+1} = s']$$

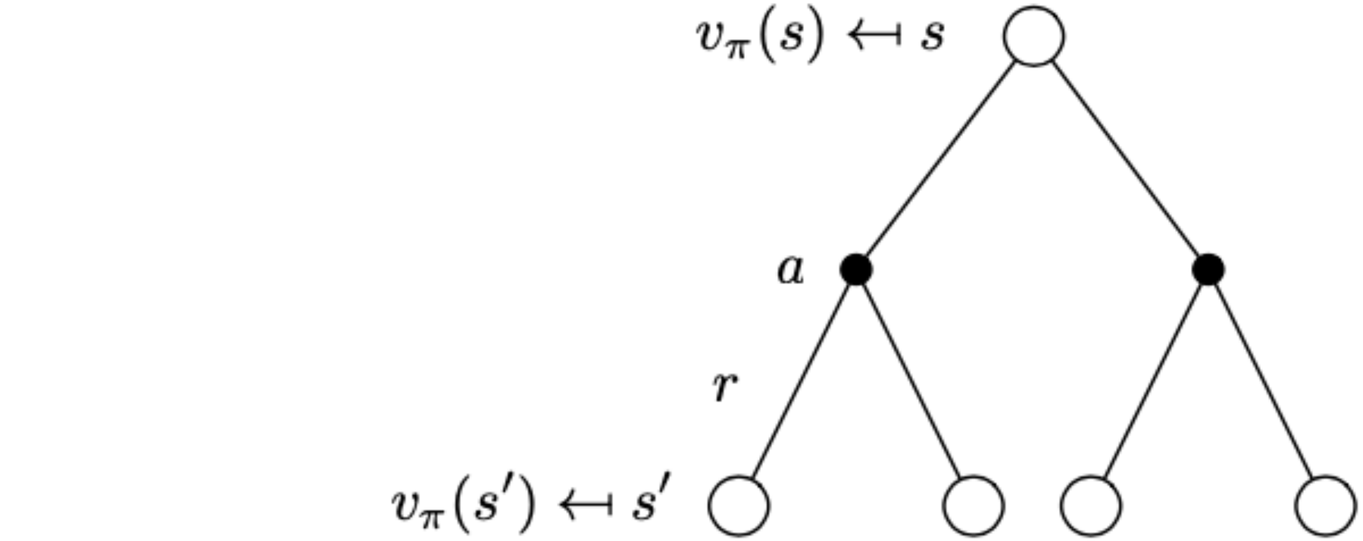
$$= r_\pi(s,a) + \gamma \sum_{s' \in \mathcal{S}} p(s' \,|\, s,a)\ \mathbf{E}_\pi[G_{t+1} \,|\, S_{t+1} = s']$$

$$= r_\pi(s,a) + \gamma \sum_{s' \in \mathcal{S}} p(s' \,|\, s,a)\ v_\pi(s')$$

Or

$$
\begin{aligned}
q_\pi(s,a) &:= \mathbf{E}_\pi[G_t \,|\, S_t = s, A_t = a] \\
&= \mathbf{E}_\pi[R_{t+1} \,|\, S_t = s, A_t = a] + \gamma \mathbf{E}_\pi[G_{t+1} \,|\, S_t = s, A_t = a] \\
&= r_\pi(s,a) + \gamma \sum_{s' \in \mathcal{S}} P^a_{ss'} \, \mathbf{E}_\pi[G_{t+1} \,|\, S_t = s, A_t = a, S_{t+1} = s'] \\
&= r_\pi(s,a) + \gamma \sum_{s' \in \mathcal{S}} P^a_{ss'} \, \mathbf{E}_\pi[G_{t+1} \,|\, S_{t+1} = s'] \\
&= r_\pi(s,a) + \gamma \sum_{s' \in \mathcal{S}} P^a_{ss'} \, v_\pi(s')
\end{aligned}
$$

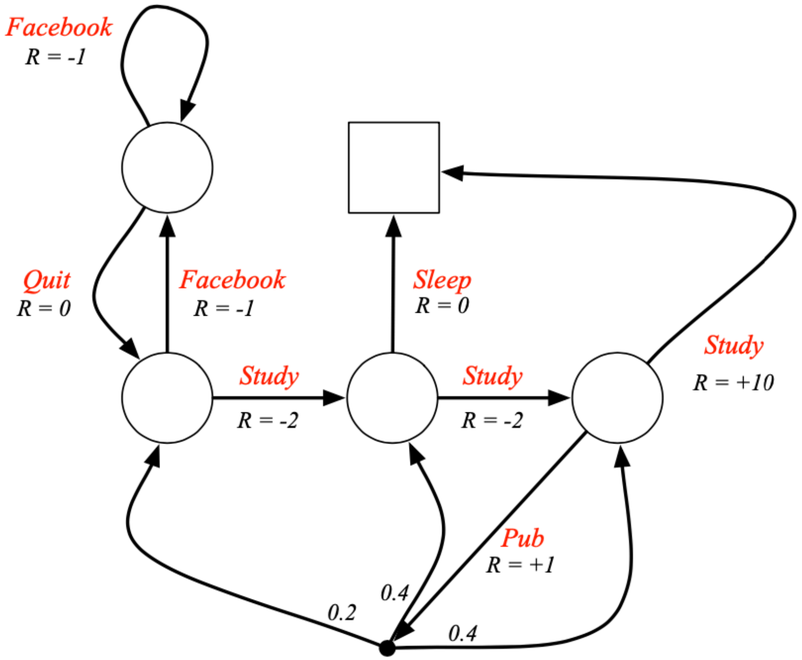
Bellman Expectation Equation for $v_\pi(2)$

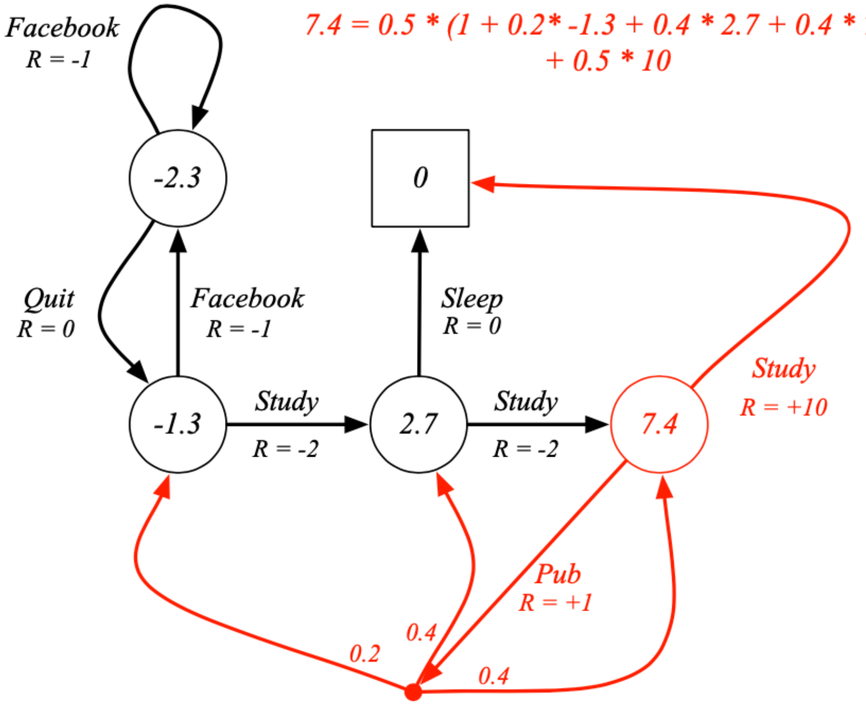


$$v_\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) \left(\mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a v_\pi(s') \right)$$

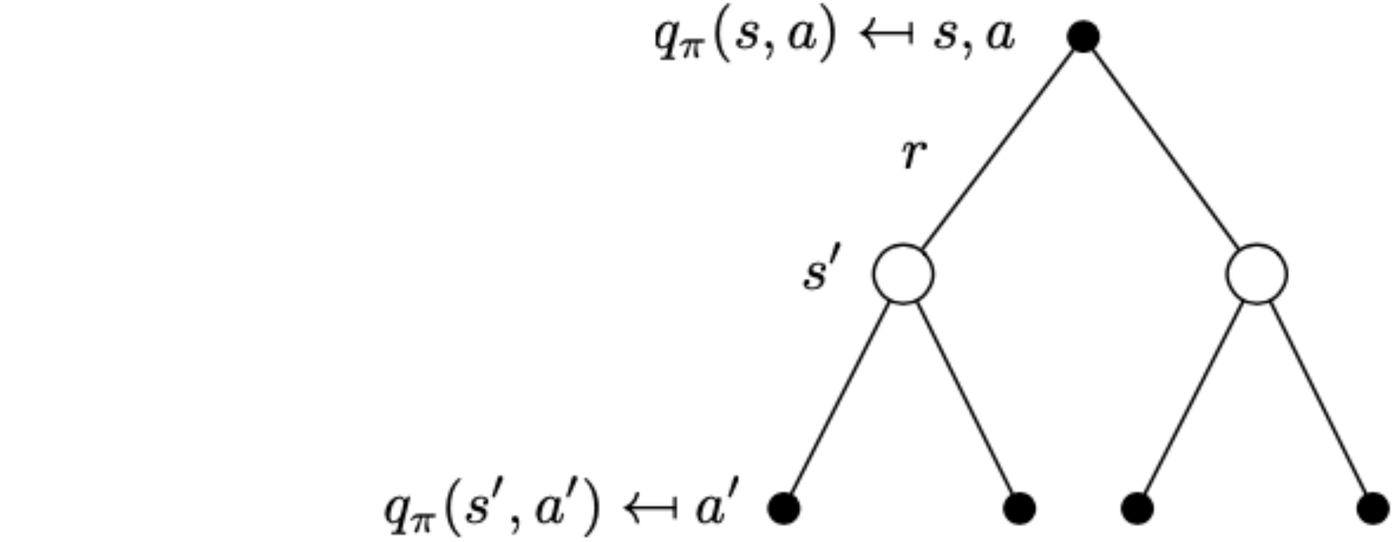
Example: Bellman Expectation Equation in Student MDP

$$v_\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) \left(\mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a v_\pi(s') \right)$$





Bellman Expectation Equation for $q_\pi(2)$



$$q_\pi(s,a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a \sum_{a' \in \mathcal{A}} \pi(a'|s') q_\pi(s',a')$$

Bellman Expectation Equation in matrix form

Matrix form

$$v_\pi = \mathcal{R}^\pi + \gamma \mathcal{P}^\pi v_\pi$$

Direct solution

$$v_\pi = (I - \gamma \mathcal{P}^\pi)^{-1} \mathcal{R}^\pi$$

Optimal Value Function

Definition

The *optimal state-value function* $v_*(s)$ is the maximum value function over all policies

$$v_*(s) = \max_\pi v_\pi(s)$$

The *optimal action-value function* $q_*(s, a)$ is the maximum action-value function over all policies

$$q_*(s, a) = \max_\pi q_\pi(s, a)$$

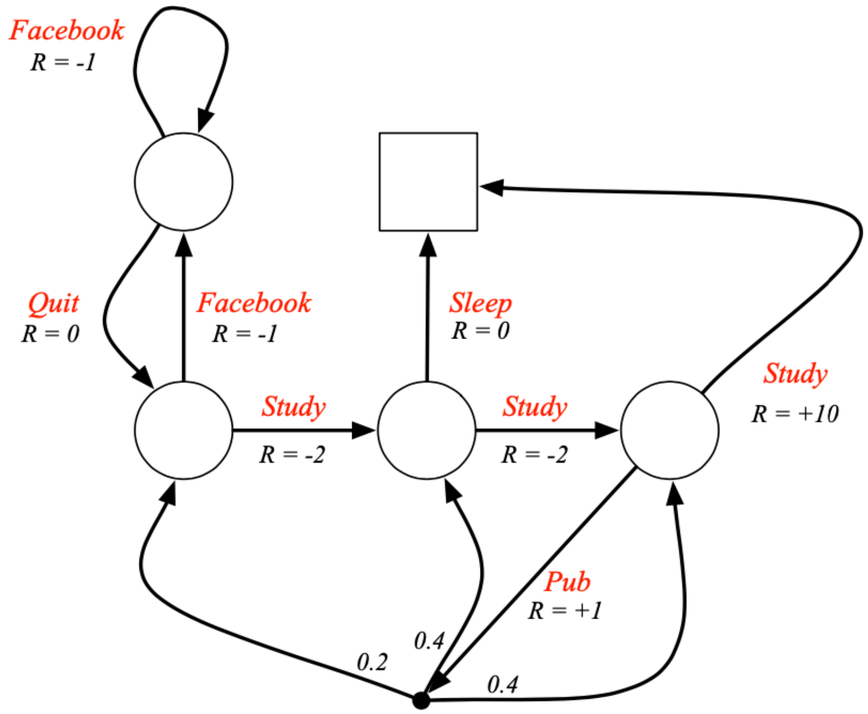
Remark

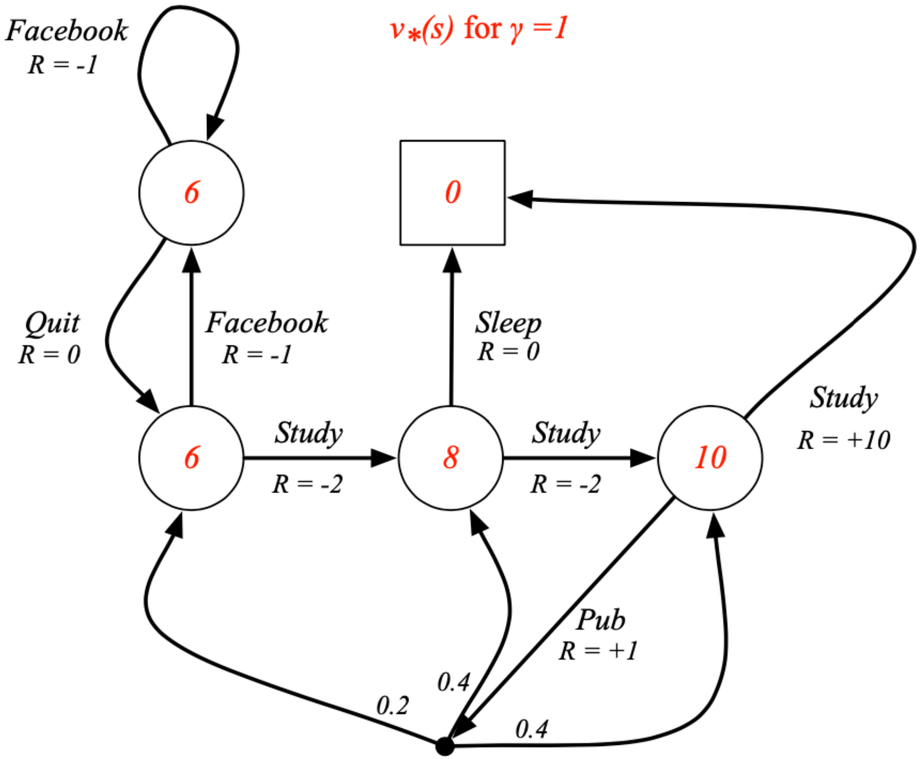
- The optimal value function specifies the best possible performance in the MDP
- An MDP is “solved” when we know the optimal value function.
- It remains to find an explicit optimal policy

Example: Optimal Value Function for Student MDP

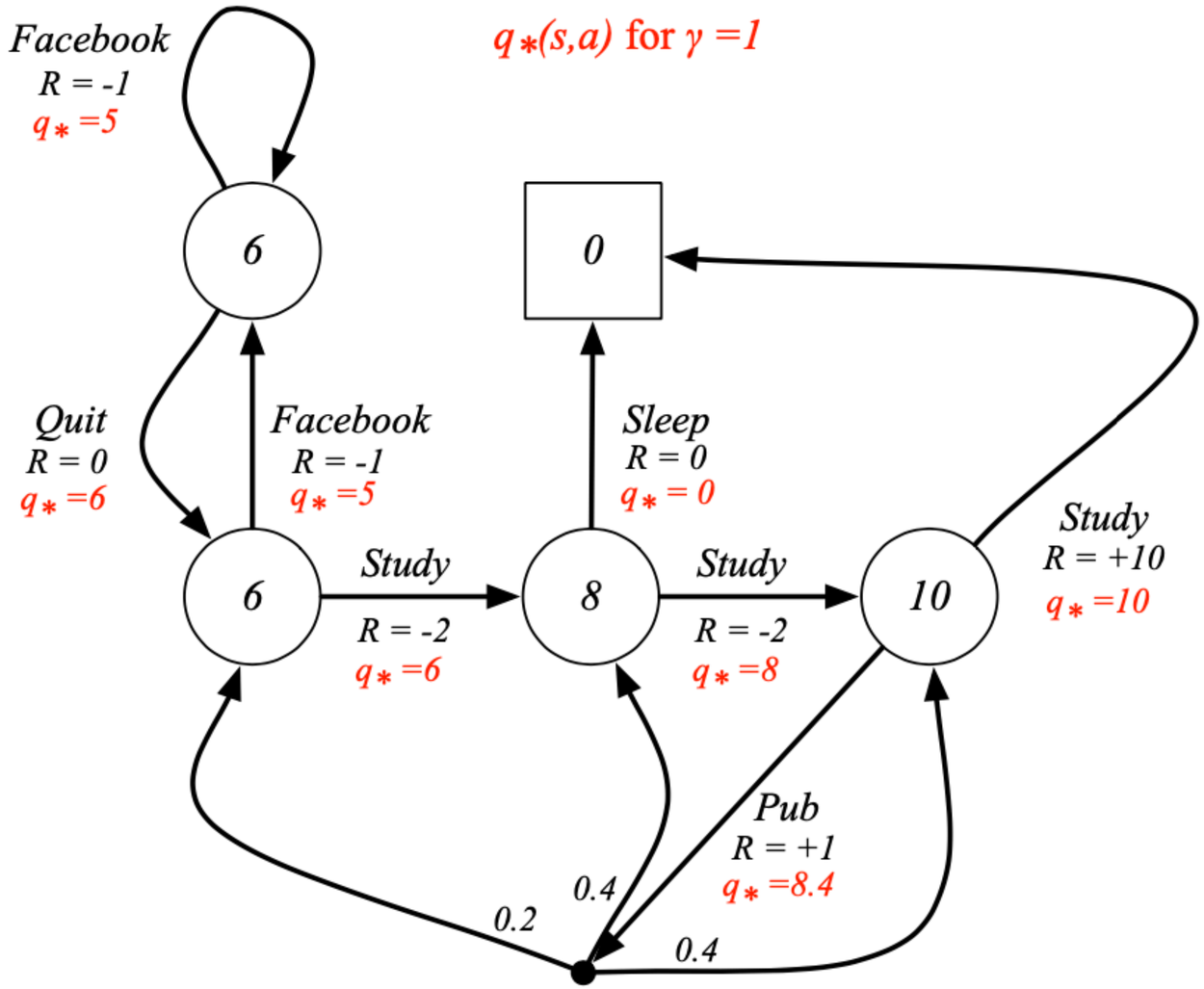
$$v^*(s) = ?$$

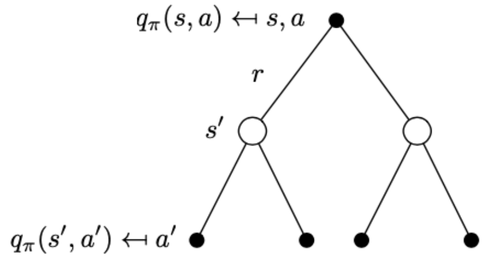
$$v_*(s) = \max_\pi v_\pi(s)$$





Example: Bellman Expectation Equation in Student MDP





$$q_\pi(s,a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a \sum_{a' \in \mathcal{A}} \pi(a'|s') q_\pi(s',a')$$

Optimal Policy

Define a partial ordering over policies

$$\pi \geq \pi' \text{ if } v_\pi(s) \geq v_{\pi'}(s), \forall s$$

Theorem

For any Markov Decision Process

- *There exists an optimal policy π_* that is better than or equal to all other policies, $\pi_* \geq \pi, \forall \pi$*
- *All optimal policies achieve the optimal value function, $v_{\pi_*}(s) = v_*(s)$*
- *All optimal policies achieve the optimal action-value function, $q_{\pi_*}(s, a) = q_*(s, a)$*

pai exist but not need to be unique

Finding an optimal policy

An optimal policy can be found by maximising over $q_*(s, a)$,

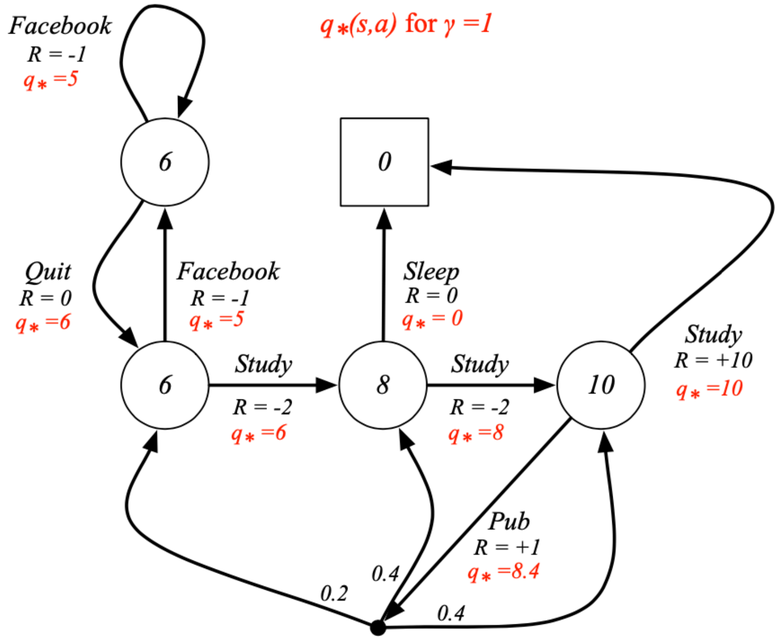
$$\pi_*(a|s) = \begin{cases} 1 & \text{if } a = \underset{a \in \mathcal{A}}{\text{argmax}}\ q_*(s, a) \\ 0 & \textit{otherwise} \end{cases}$$

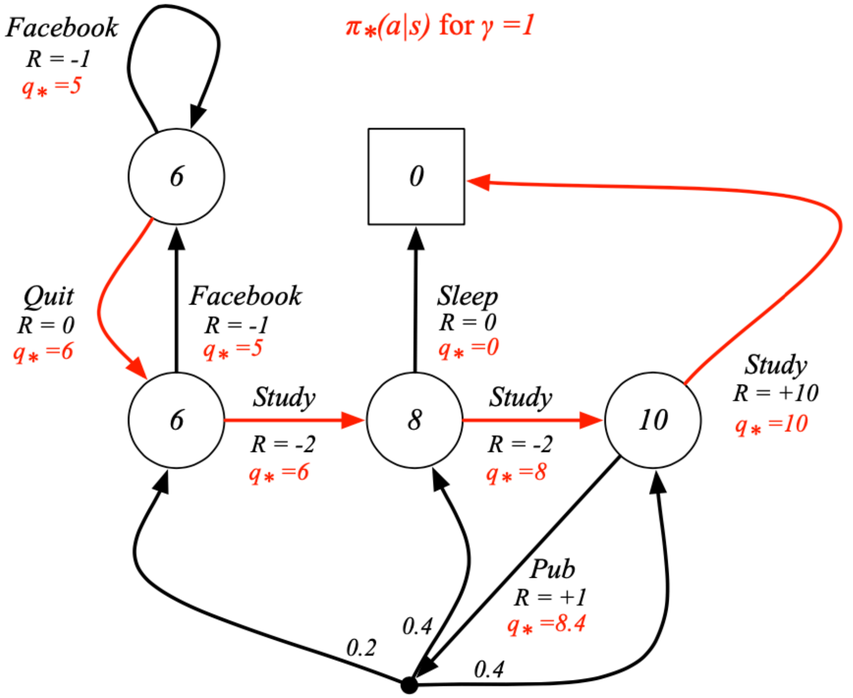
- There is always a deterministic optimal policy for any MDP
- If we know $q_*(s, a)$, we immediately have the optimal policy

Optimal policy: (1) deterministic/random; (2) maybe many

Example: Optimal policy for Student MDP

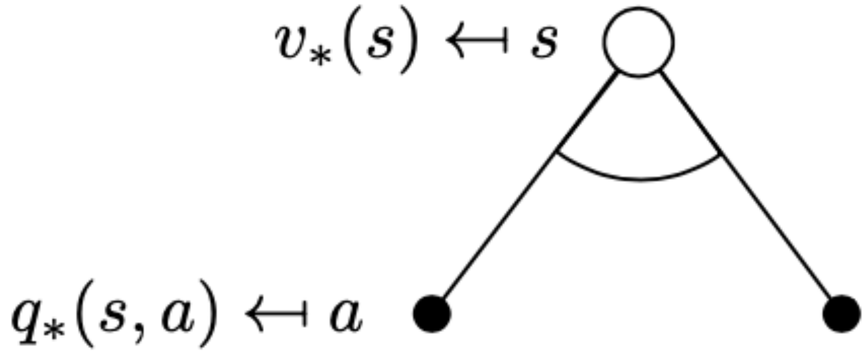
Optimal Policy π^*?





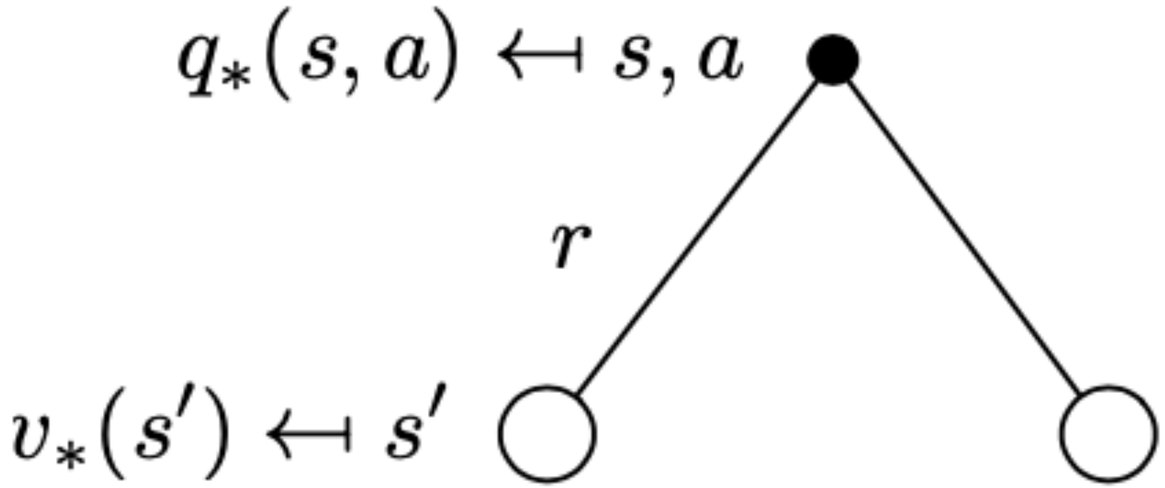
Bellman Optimality Equation for v^*

The optimal value functions are recursively related by the Bellman optimality equations:



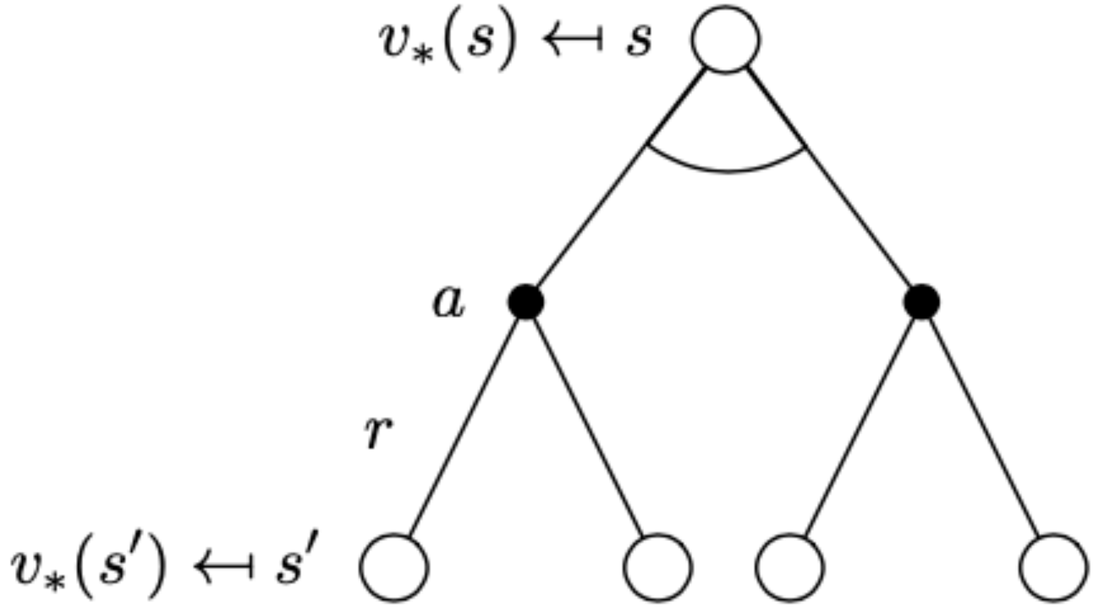
$$v_*(s) = \max_a q_*(s,a)$$

Bellman Optimality Equation for q^*



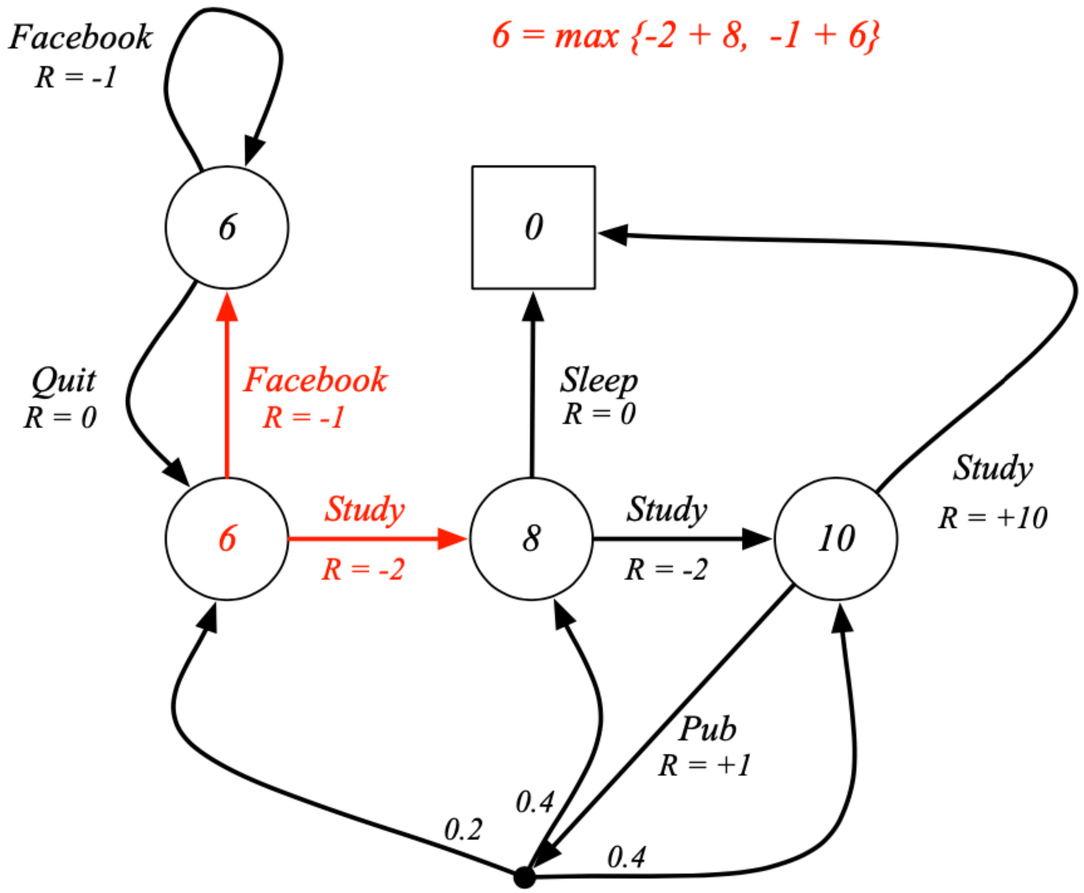
$$q_*(s, a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a v_*(s')$$

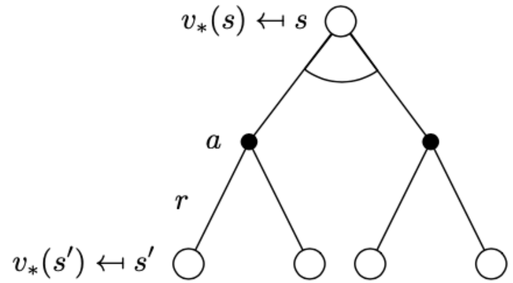
Bellman Optimality Equation for $v^*(2)$



$$v_*(s) = \max_a \left(\mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a v_*(s') \right)$$

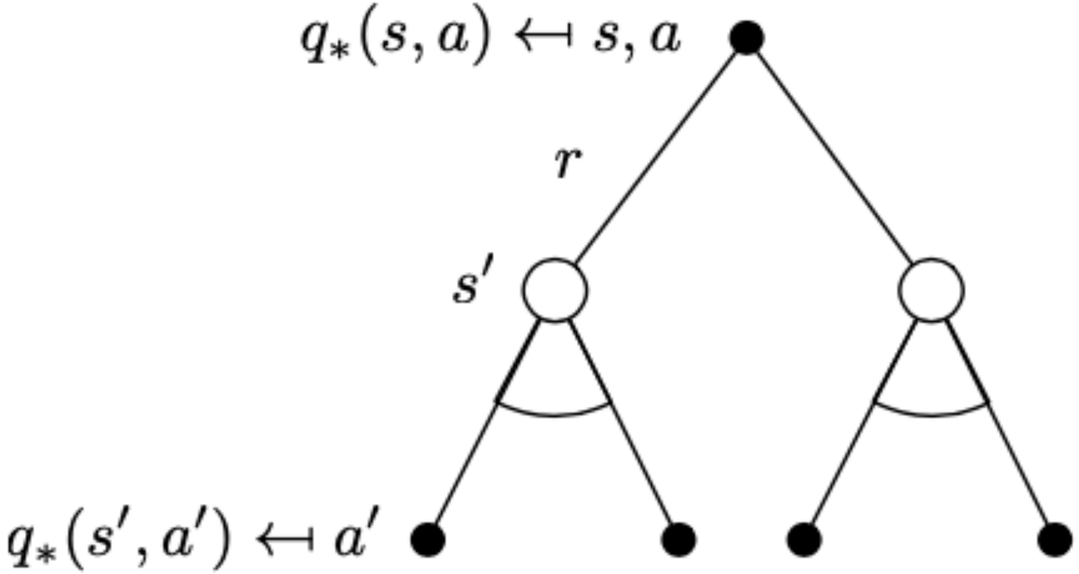
Example: Bellman Optimality Equation in Student MDP





$$v_*(s) = \max_a \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a v_*(s')$$

Bellman Optimality Equation for $q^*(2)$



$$q_*(s,a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a \max_{a'} q_*(s',a')$$

Solving the Bellman Optimality Equation

- Bellman Optimality Equation is non-linear
- No closed form solution (in general)
- Many iterative solution methods
  - Value Iteration
  - Policy Iteration
  - Q-learning
  - Sarsa

